THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **The Human Eye and the Colourful World** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The image in the human eye is formed on the 1

(a) pupil **(b)** retina **(c)** cornea **(d)** iris

2. Myopia is corrected by using 1

(a) a convex lens **(b)** a cylindrical lens

(c) a plane mirror **(d)** a concave lens

3. Splitting of white light into its seven colours is called 1

(a) scattering **(b)** dispersion **(c)** refraction **(d)** reflection

4. Most of the refraction of light entering the eye takes place at the 1

(a) eye lens **(b)** iris

(c) retina **(d)** outer surface of the cornea

5. Hypermetropia is corrected by using 1

(a) a convex lens **(b)** a concave lens **(c)** a prism **(d)** a concave mirror

6. Which colour bends least on passing through a glass prism? 1

(a) Blue **(b)** Green **(c)** Red **(d)** Violet

7. The size of the pupil is controlled by the 1

(a) retina **(b)** ciliary muscles **(c)** iris **(d)** optic nerve

8. In myopia, the image of a distant object is formed 1

(a) in front of the retina **(b)** on the iris

(c) behind the retina **(d)** on the retina

9. Which colour bends most on passing through a prism? 1

(a) Yellow **(b)** Violet **(c)** Red **(d)** Orange

10. The ability of the eye lens to adjust its focal length is called 1

(a) myopia **(b)** dispersion **(c)** accommodation **(d)** scattering

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by the power of accommodation of the eye? 2

12. What are the causes of myopia? 2

13. What is the Tyndall effect? 2

14. Name the part of the eye which controls the amount of light entering and its working. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the structure and function of the cornea, iris, pupil and retina. 3

16. Why is it difficult for older persons to see nearby objects? 3

17. Why do planets not twinkle? 3

OR

How do the ciliary muscles help in focusing on near and far objects? 3

18. What is a cataract? How is it treated? 3

OR

What is scattering of light? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Draw the structure of the human eye (describe) and explain the function of its parts: cornea, iris, pupil, lens, retina, optic nerve. 5

OR

What is dispersion of light? Describe Newton's experiment with two prisms. 5

20. What is accommodation? Explain how the eye focuses on objects at different distances and mention the near and far points. 5

OR

Explain the formation of a rainbow with the help of a description of the path of light in a water droplet. 5

END OF QUESTION PAPER — Total: 40 Marks

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